

Phase 2: Requirement Analysis

1. Introduction

The Requirement Analysis phase defines the functional, non-functional, and data requirements for the **Human Development Index (HDI) Prediction System**. The system uses Machine Learning techniques to predict a country's Human Development Index category based on key socio-economic indicators, enabling users to analyze development levels quickly and accurately.

2. Functional Requirements

These define what the system should do:

Data Input: Users must be able to manually enter the following indicators through a web interface:

- Life Expectancy
- Mean Years of Schooling
- Expected Years of Schooling
- Gross National Income (GNI) per Capita (PPP)

Prediction Generation: The system must process the input data using a pre-trained Machine Learning model to predict the HDI category.

Result Display: The application must display the predicted **HDI Score, HDI Category (Very High, High, Medium, or Low)**, and a brief interpretation of the result.

User-Friendly Interface: The system should provide an intuitive and responsive interface for easy interaction by users with no programming knowledge.

Prediction History (Future/Optional): The application may store previous predictions for future reference and analysis.

3. Non-Functional Requirements

These define how the system should perform:

Accuracy: The Machine Learning model should maintain a high prediction accuracy (preferably above 95%).

Usability: The web application should be simple, responsive, and accessible on desktop and mobile devices.

Performance: The system should generate predictions within **2 seconds** of receiving user input.

Reliability: The application should consistently produce accurate predictions with minimal errors.

Deployment: The application should be compatible with cloud platforms such as **Render, Heroku**, or similar hosting services.

4. Data Requirements

Data Source: Human Development Index (HDI) Dataset containing country-wise development indicators.

Features Required:

- Life Expectancy
- Mean Years of Schooling
- Expected Years of Schooling
- Gross National Income (GNI) per Capita (PPP)

Target Variable:

- Human Development Index (HDI) Score
 - HDI Category (**Very High, High, Medium, or Low**)
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5. Software & Hardware Requirements

Programming Language: Python 3.x

Libraries: Pandas, NumPy, Scikit-learn, Flask, Pickle, Matplotlib

Frontend: HTML5, CSS3, Bootstrap (Optional)

Development Environment: Jupyter Notebook (for data preprocessing and model training), Visual Studio Code (VS Code) for application development

Machine Learning Algorithm: Random Forest Regressor (or the model used in your project)

Operating System: Windows 10/11, Linux, or macOS

Hardware Requirements:

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB (8 GB Recommended)
- Storage: 500 MB free disk space or more
- Internet Connection: Required for deployment and online access (optional)

This follows the exact structure of your sample while being tailored to your **Human Development Index (HDI) Prediction System**.